

Enhanced indexation with stochastic dominance

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Abstract

Keywords:

Hi John, hopefully this and long relatively informal document can be a good introduction to the area, to the state-of-the-art and to some research ideas. I took some time to add several references, both on stochastic dominance and on enhanced indexation.

1. Introduction

An enhanced indexation optimisation approach must define what it means by “enhanced” - what is the characteristic of the benchmark index that it aims to improve. Several papers in literature have dealt with this problem by approaching it in different ways.

To the best of my knowledge, the most common approach in enhanced indexation is to attempt to outperform a benchmark index while constraining either the tracking error or relative underperformance (relative risk), so trying to balance to conflicting objectives. For reference, a evaluation of funds of this type was presented by Ahmed & Nanda (2005).

In this category, Wu et al. (2007) proposed a model to maximise the information ratio, which is the ratio between excess return and tracking error. Canakgoz & Beasley (2009) maximise the regression intercept between portfolio and index returns, while keeping the slope as close to 1 as possible. Li et al. (2011) proposed a multiobjective model to maximise excess return and minimise negative tracking error (below the index). Lejeune (2012) proposed a game-theoretical approach to maximise excess return while constraining maximum tracking error (referred to as relative risk). Mezali & Beasley (2013) employ quantile regression, which allows to easily switch between index tracking and enhanced indexation within the same model. Guastaroba & Speranza (2012) used the absolute deviation form of the tracking error, instead of the quadratic one. Bruni et al. (2015) maximise the excess return while limiting the maximum underperformance. Guastaroba et al. (2016) tries to balance risk and reward in enhanced indexation by optimising the Omega ratio.

Other approaches in enhanced indexation adopt different policies of what enhanced means. Meade & Beasley (2011) examine asset outperformance momentum in order to select (hopefully) outperforming portfolios. Goel et al. (2018) propose a two-step procedure, one that initially filters assets based on Hidden Markov Chains and then solves an enhanced indexation model based

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on the STARR ratio. Sehgal & Mehra (2019) present an enhanced indexation model based on weighted CVaR - in this paper, they compare their model to several others, including the stochastic dominance one that is the basis for this document (Roman et al., 2013). Beraldi & Bruni (2020) propose a chance-constraint based formulation in which eventual underperformance is controlled probabilistically.

Apart from Roman et al. (2013), there have also been enhanced indexation approaches based on stochastic dominance. I included them later in this text.

Stochastic dominance (SD) is a different approach when compared to the more traditional approaches in enhanced indexation. In the other approaches, we usually have a clear view that the resulting model should have limited “extra” risk when compared to the benchmark, and excess return. We have then to somehow define these two measures, and generally the distribution of returns is summarised in a couple of statistics, used to rank portfolio choices. These statistics often ignore important aspects of the distributions, such as fat tails and skewness.

The SD relationship, compares the entirety of the portfolios distributions, thus preventing the potential problem of focusing on specific statistics while ignoring other statistical information.

2. Stochastic dominance

SD is a partial ordering between two random variables. Let X e Y be these two variables and let $F_X(x) = \Pr(X \leq x)$ be the cumulative distribution function of X . X stochastically dominates Y in **first order** ($X \succeq_{FSD} Y$) if:

$$F_X(z) \leq F_Y(z) \quad \forall z$$

with at least one strict inequality.

In regards to portfolio selection, FSD is generally a very “strong” condition. Second-order stochastic dominance ($X \succeq_{SSD} Y$) is less restrictive, and defined as:

$$\int_{-\infty}^z F_X(w)dw \leq \int_{-\infty}^z F_Y(w)dw \quad \forall z$$

with at least one strict inequality.

Figures 1 and 2 illustrate, through cumulative distribution functions, the concept of first-order and second-order stochastic dominance, respectively (sorry for the charts in Portuguese).

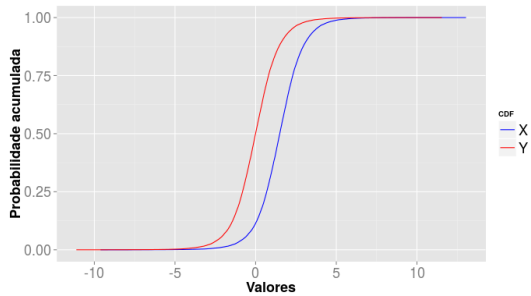


Figure 1: $X \succeq_{FSD} Y$

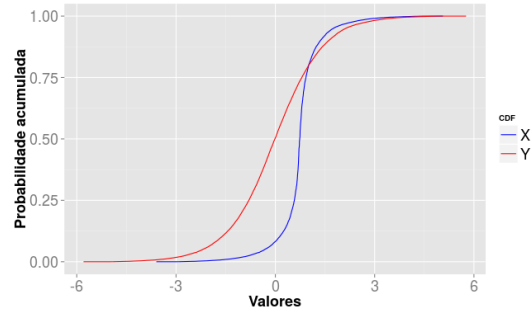


Figure 2: $X \succeq_{SSD} Y$

As an example, consider a die and three different bets A , B and C , with the following payoff table:

Die roll x	1	2	3	4	5	6
A	1	2	3	4	5	6
B	1	2	4	4	5	6
C	3	3	3	3	3	6

We can see that $B \succeq_{FSD} A$, and that neither $C \not\succeq_{FSD} A$ nor $A \not\succeq_{FSD} C$. In Figure 3 we show the cumulative distribution function of both A and C , and in Table 1 the numerical second-order stochastic dominance relation. From this example we can see that $C \succeq_{SSD} A$.

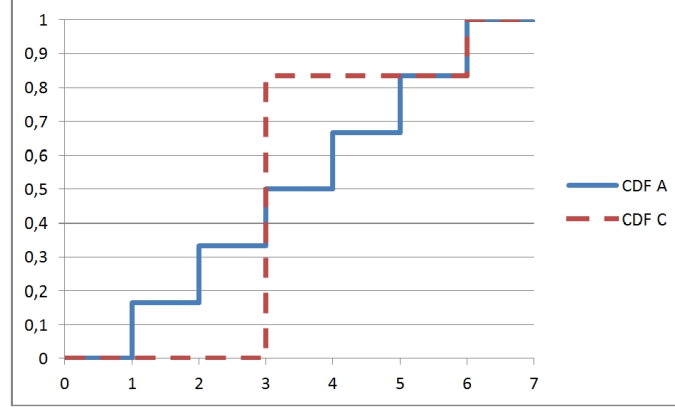


Figure 3: Cumulative distributions of A and C

Die roll x	1	2	3	4	5	6
$\int_{-\infty}^x F_A(t) dt$	0	1/6	3/6	6/6	10/6	15/6
$\int_{-\infty}^x F_C(t) dt$	0	0	0	5/6	10/6	15/6
$\int_{-\infty}^x F_A(t) - F_C(t) dt$	0	1/6	4/6	5/6	0	0

Table 1: SSD relation between C and A

One of the things that make stochastic dominance so appealing is its strong theoretical foundation. They are intimately linked to the expected utility theory of von Neumann & Morgenstern (1947). Both risk and reward measures are utility functions, and there is generally no consensus among investors of what represents an ideal utility function. Stochastic dominance eliminates the need to specify a utility function (Whitmore & Findlay, 1978): a portfolio X is preferable over portfolio Y for every investor with monotonic increasing utility functions if and only if $X \succeq_{FSD} Y$. Moreover, X is preferable over Y for every investor with monotonic increasing and strictly concave utility functions if and only if $X \succeq_{SSD} Y$.

In the study of economic behaviour, it is generally accepted that increasing and concave utility functions express the preference of risk-averse investors, which gives a theoretical justification for the importance of choosing SSD efficient portfolios. According to risk theory, every coherent risk measure (Artzner et al., 1999) is increasing and concave (that includes Conditional Value-at-Risk). In other words, this means that if $X \succeq_{SSD} Y$, then X has a better CVaR than Y for every possible confidence level.

It is possible to go deep into SD theory and its relation to risk and utility theory, as well as the beginning of portfolio optimisation with SD constraints. However, in my view, in order to work with enhanced indexation based on SD, this is not strictly necessary. We should of course have in mind that the theory gives an important background that justifies the approach. For references on more

theoretical papers, see Levy (1992); Ogryczak & Ruszczyński (2002); Dentcheva & Ruszczyński (2003); Dentcheva & Ruszczyński (2006).

In order to construct a linear programming formulation for the enhanced indexation formulation, let us first present the SD relation for discrete random variables. Let X and Y be discrete, with S equiprobable outcomes. Let $x_{(1)}, x_{(2)}, \dots, x_{(S)}$ be the ordered outcomes of X , from smallest to largest. The FSD and SSD relations are given by:

$$X \succeq_{FSD} Y \quad \text{if} \quad x_{(s)} \geq y_{(s)} \quad \forall s = 1, \dots, S, \quad \text{with at least one strict inequality.}$$

$$X \succeq_{SSD} Y \quad \text{if} \quad \sum_{z=1}^s x_{(z)} \geq \sum_{z=1}^s y_{(z)} \quad \forall s = 1, \dots, S, \quad \text{with at least one strict inequality.}$$

A numerical example of the calculation of the second-order stochastic dominance relation follows. Consider that we have 5 days with historical data for both X and Y :

Day	X	Y
1	3%	2%
2	-5%	3%
3	2%	2%
4	5%	-2%
5	1%	1%

With the sample data above, we first sort the returns (omitting the percentage sign):

Order	X	Y
(1)	-5	-2
(2)	1	-1
(3)	2	1
(4)	3	3
(5)	5	6

Then we calculate their cumulative sum: $X = (-5, -4, -2, 1, 6)$ and $Y = (-2, -1, 1, 3, 6)$. If we subtract the cumulative sum of ordered returns of X from Y , we get $(3, 3, 3, 2, 0)$. Since all numbers are nonnegative, with at least one positive number, we say that $Y \succeq_{SSD} X$.

In the linear programming formulation of Roman et al. (2013), the goal is to find the portfolio that maximises the minimum value in the SSD relation with regards to a benchmark. For instance, if the sorted values of $Y_2 = (-4, 1, 2, 3, 5)$, the SSD relation between Y_2 and X is $(1, 1, 1, 1, 1)$. According to this formulation, Y_2 is preferable over Y as the minimum value in the SSD relation is 1 as opposed to 0.

3. Enhanced indexation formulation

Finding a portfolio that dominates the benchmark involves finding weights such that the cumulative ordered returns are higher than those of the benchmark at every point of the distribution. At first, this problem was modelled as a multi-objective formulation by Roman et al. (2006).

Later on, Fábíán et al. (2011a) proposed a cutting-plane reformulation of the SSD model - one with an exponential number of constraints. The reformulation itself was inspired by a cutting-plane reformulation of the CVaR optimisation model by Küenzi-Bay & Mayer (2006). In the SSD reformulation, the multiple objectives are converted into one were we want to maximise the

minimum value in the SSD relation. In that paper, the authors proposed a very simple separation algorithm, basically by inspection - I will outline how it works below.

Finally, Roman et al. (2013) adapted the model proposed by Fábíán et al. (2011a) into an enhanced indexation problem. Let:

N	number of assets into which we may invest.
C	total amount of cash to invest.
$R = (R_1, \dots, R_n)^T$	discrete random variables denoting the returns of assets in $\{1, \dots, N\}$.
R^I	discrete random variable denoting the returns of the benchmark
S	number of scenarios
r_{is}	return of asset i under scenario $s = \{1, \dots, S\}$
r_a^I	return of index under scenario s
$\text{Tail}_\alpha(R)$	unconditional expectation of the least $\alpha \times 100\%$ outcomes of R

In the case of equiprobable scenarios, Fábíán et al. (2011a) defined, in Eq (10) of their paper, for a portfolio p , $\text{Tail}_{\frac{s}{S}}(R^p)$ as:

$$\text{Tail}_{\frac{s}{S}}(R^p) = \frac{1}{S} \sum_{j=1}^s r_{p(j)}$$

where $r_{p(j)}$ denotes the j -th worst return of the portfolio. For equiprobable scenarios:

$$\text{CVaR}_{\frac{s}{S}}(R^p) = - \sum_{j=1}^s \frac{1}{s} r_{p(j)}$$

for $\text{CVaR}_{\frac{s}{S}}$ with confidence level $\frac{s}{S}$. Fábíán et al. (2011a) (and others) showed that all three statements below are equivalent:

$$\begin{aligned} R^p &\succeq_{SSD} R^I \\ \text{CVaR}_{\frac{s}{S}}(R^p) &\leq \text{CVaR}_{\frac{s}{S}}(R^I) & \forall s = \{1, \dots, S\} \\ \text{Tail}_{\frac{s}{S}}(R^p) &\geq \text{Tail}_{\frac{s}{S}}(R^I) & \forall s = \{1, \dots, S\} \end{aligned}$$

Let $\hat{\tau} = (\hat{\tau}_1, \dots, \hat{\tau}_S) = (\text{Tail}_{\frac{1}{S}} R^I, \dots, \text{Tail}_{\frac{S}{S}} R^I)$. Let w_i be the proportion to be invested in asset $i = 1, \dots, N$ and let $\mathcal{V}$ be an auxiliary variable. The formulation that maximises the minimum value in the SSD relation is given by:

$$\begin{aligned} \max \quad & \mathcal{V} \\ \text{subject to} \quad & \mathcal{V} \leq \frac{1}{S} \sum_{j=1}^s \sum_{i=1}^N r_{i(j)} w_i - \hat{\tau}_s \quad s = 1, \dots, S \\ & \sum_{i=1}^N w_i = 1 \\ & \mathcal{V} \in \mathbb{R} \end{aligned}$$

The obvious problem with the formulation above is that we do not know the order of the portfolio returns prior to choosing w_i . Inspired by the CVaR cutting-plane reformulation by Küenzi-Bay & Mayer (2006), Fábíán et al. (2011a) proposed the following redefinition of $\text{Tail}_{\frac{s}{S}}(R^p)$:

$$\text{Tail}_{\frac{s}{S}}(R^p) = \frac{1}{S} \min \sum_{j \in \mathcal{J}} \sum_{i=1}^N r_{ij} w_i \quad \text{such that } \mathcal{J} \subset \{1, \dots, S\}, |\mathcal{J}| = s.$$

We then reformulate the model above as:

$$\max \quad \mathcal{V} \tag{1}$$

$$\text{subject to} \quad \mathcal{V} \leq \frac{1}{S} \sum_{j \in \mathcal{J}_s} \sum_{i=1}^N r_{ij} w_i - \hat{\tau}_s \quad \forall \mathcal{J}_s \subset \{1, \dots, S\}, \tag{2}$$

$$|\mathcal{J}_s| = s, s = \{1, \dots, S\}$$

$$\sum_{i=1}^N w_i = 1 \tag{3}$$

$$\mathcal{V} \in \mathbb{R} \tag{4}$$

The formulation above finds the portfolio that maximises the minimum value in the SSD relation. If none exists, the objective function value is negative and represents the portfolio that comes the closest to dominating the benchmark.

In order to solve this formulation, Fábíán et al. (2011a) proposed a very simple separation algorithm:

- (1) Start with at least one Constraint of type (2) to prevent the formulation from being unbounded.
- (2) Solve the formulation to obtain portfolio $w^* = (w_1^*, \dots, w_N^*)$ with returns $r_s^p = \sum_{i=1}^N r_{is} w_i^*$.
- (3) Sort the portfolio returns and, by inspection, check if $\mathcal{V} > \sum_{j=1}^s r_{(j)}^p$ for any $s = \{1, \dots, S\}$. Add the corresponding violated constraint(s) to the model and go back to Step (2).

From experience, I found this model to be quite easy to solve in practice. Sections 3 and 4 in Roman et al. (2013) provide a good, readable and more complete summary of everything I wrote.

A slightly enhanced version of the formulation above was proposed by Fábíán et al. (2011b). The modified formulation is given by:

$$\max \quad \mathcal{V} \tag{5}$$

$$\text{subject to} \quad \frac{s}{S} \mathcal{V} \leq \frac{1}{S} \sum_{j \in \mathcal{J}_s} \sum_{i=1}^N r_{ij} w_i - \hat{\tau}_s \quad \forall \mathcal{J}_s \subset \{1, \dots, S\}, \tag{6}$$

$$|\mathcal{J}_s| = s, s = \{1, \dots, S\}$$

$$\sum_{i=1}^N w_i = 1 \tag{7}$$

$$\mathcal{V} \in \mathbb{R} \tag{8}$$

The only difference is that the tails for different s are scaled. Fábíán et al. (2011b) justified this approach by making a theoretical connection to risk minimisation theory. Their results section suggest that the portfolio distribution obtained by solving the scaled version generally lies to the

right of the distribution obtained with the unscaled version. Indeed, from experience, overall the out-of-sample results with the scaled model are better - not always, but generally.

The scaled model gives more importance to the right tail (extreme positive returns) than the unscaled model does. In the unscaled model, the contribution of the right tail in the SSD relation is “diluted” among all the previous ordered returns. The scaled formulation tends to find portfolios with more prominent right tails, although also generally more risky. In Section 5 I included two charts to show the difference.

In order to easily add other real-world constraints, such as limited short-selling or cardinality, I implemented a Branch-and-cut version of this formulation. We published this in Valle et al. (2017). In that paper we also studied how to reshape/improve the reference distribution (the benchmark). The idea was that a portfolio that dominates a benchmark with more desirable statistical properties is more likely to have better out-of-sample performance. With regards to computational performance, we are able to solve the problem quite quickly for most cases - I have tested up to 5700 assets and 250 scenarios, or up to 500-600 assets and 1200 scenarios. The most notable exception is the addition of cardinality constraints coupled with buy-in thresholds, which makes the problem harder.

4. Other approaches to SSD

There are other different linear programming approaches to SSD. One of these approaches is that of Post & Kopa (2013); Kopa & Post (2015). The original idea was not necessarily an enhanced indexation model, but rather a test of SSD efficiency. In Kopa & Post (2015), the authors developed a primal-dual linear programming framework to evaluate whether a fixed portfolio is SSD-efficient with respect to all feasible portfolios in a given polytope. The authors showed that the dual of their formulation can select, if it exists, another portfolio (belonging to the polytope) that SSD-dominates the fixed portfolio. In this case, the dual works as an enhanced indexation model. I have read this paper but in truth I need to go back to it to fully understand their model and, if necessary, implement it.

Quite recently, Cesarone et al. (2022) used our benchmark reshaping approach (Valle et al., 2017) to compare the out-of-sample performance of both the model here presented (Roman et al., 2013) and that of Kopa & Post. The authors found the Roman et al. model more flexible in the sense that if no feasible portfolio exists, the Kopa & Post model fails, while Roman et al. is able to find the one that comes the closest to dominating the benchmark. This is an important case as when the benchmark is reshaped it is less likely that a dominating portfolio exists.

While my suggestion is to work on the Roman et al. model, having a better understanding of the Kopa & Post model might be useful since it is a more general approach. It was built for any order, not only the second (the third order SD, for instance, requires a double integral of the cumulative distribution function).

Luedtke (2008) presented a model that includes FSD constraints based on a cutting-plane formulation for problems with integrated chance constraints (Haneveld & Van Der Vlerk, 2006). Their model is integer, but relaxing integrality yields a formulation with SSD constraints. Their objective is to maximise portfolio expected return, subject to these constraints. In this paper they generalise a first linear-programming structure for SSD proposed by Kuosmanen (2004). I have not gone into detail in this paper - it has a more theoretical build-up.

Another optimisation approach with SD was developed by Bruni et al. (2012). Their approach is specifically for enhanced indexation. They use a general definition of stochastic dominance and propose a criterion called “Cumulative Zero-order stochastic ϵ -dominance” (CZS ϵ D). Zero-order SD happens when all returns of a portfolio are superior to all other returns of the other portfolio, irrespective of the ordering. The authors attempt to minimise underperformance by adding an exponential number of constraints related to the CZS ϵ D criterion, where ϵ is the maximum

underperformance allowed. The same model was reintroduced in a more recent and detailed paper (Bruni et al., 2017). The separation algorithm is equivalent to optimising CVaR (via linear programming).

Bruni et al. (2017) also includes a comprehensive testing framework, in which they implemented several (Luedtke, 2008), (Bruni et al., 2012), (Roman et al., 2013) and (Kopa & Post, 2015). They also implemented another formulation which I am yet to review (Lizyayev & Ruszczyński, 2012), based on a concept called “almost stochastic dominance”. Their conclusion is that the models are all far from ideal and hard to solve, in favor of their “approximate SD” approach. This contradicts my own experience.

Something I found odd was that one of the authors of Bruni et al. (2017), Francesco Cesarone, is the main author of the recent comparative study (Cesarone et al., 2022), and he did not use his own model in the study. In fact, he implemented only Kopa & Post (2015) and Roman et al. (2013), with a final view more favourable to Roman et al..

Another application of SSD to enhanced indexation was that by Hodder et al. (2015). They implemented Kuosmanen (2004) and Kopa & Post (2015), with clear descriptions of the models (if I am to study these models deeper, I will probably look into this work). The Kopa & Post requires arbitrary values for certain parameters, and the authors in this paper proposed a way to assign these values. They focus on constructing multiple portfolios that SSD-dominate the benchmark, and choose one or a few among them. The parameter definition proposed here was used by Cesarone et al. (2022) when implementing the Kopa & Post formulation.

Sharma et al. (2017) used a relaxed-SSD model for enhanced indexation. They constructed their paper based on two approaches, Lizyayev & Ruszczyński (2012) and Dentcheva & Ruszczyński (2003); Dentcheva & Ruszczyński (2006). I am yet to review these papers.

Goel & Sharma (2021) construct an approach more focused on mean return, and based on both Dentcheva & Ruszczyński (2003) and Roman et al. (2013). I am also yet to review these papers.

While there are several other approaches to SSD and enhanced indexation, and several different models, none of the base models seem to be as simple and flexible as Roman et al. (2013). I stand to be corrected though, as I read them more carefully. My suggestions for future work are then on top of Roman et al.. That is not to say that I could definitely learn and find new ideas from the works above, so I intend to go over them properly. Bruni et al. (2017) and Hodder et al. (2015) can be very helpful in this case.

5. Ideas for research

There are generally many portfolios that dominate a benchmark with regards to SSD, and an investor must decide which to select. The two models shown above, scaled and unscaled, pick very different portfolios. The unscaled formulation maximises the minimum value of the SSD relationship, while the scaled formulation maximises the scaled minimum. In Figures 4 and 5 I plotted the FSD and SSD relation between an in-sample portfolio selected with SP500 assets, and the SP500 index itself. The portfolios were selected with 500 in-sample days ending in 01-01-2022, so including the COVID drop.

Both dominate the benchmark, but the unscaled portfolio has a peak around the middle. The descending curve in SSD means a negative value in FSD. One advantage of the unscaled portfolio is that its left tail is much higher than that of the benchmark - the worst return is 10.62% higher than that of the benchmark. The scaled portfolio is more risk-prone as it gives more emphasis on the right tail. It almost completely dominates the benchmark with regards to FSD, but it falls short in the extreme right tail. Sometimes we do find scaled portfolios that dominate the benchmark with regards to FSD as well. A potential problem with the scaled model is that the domination in the left tail is less pronounced - in the example, the worst return is 6.9% higher. In these examples, both portfolios have a worse extreme right tail.

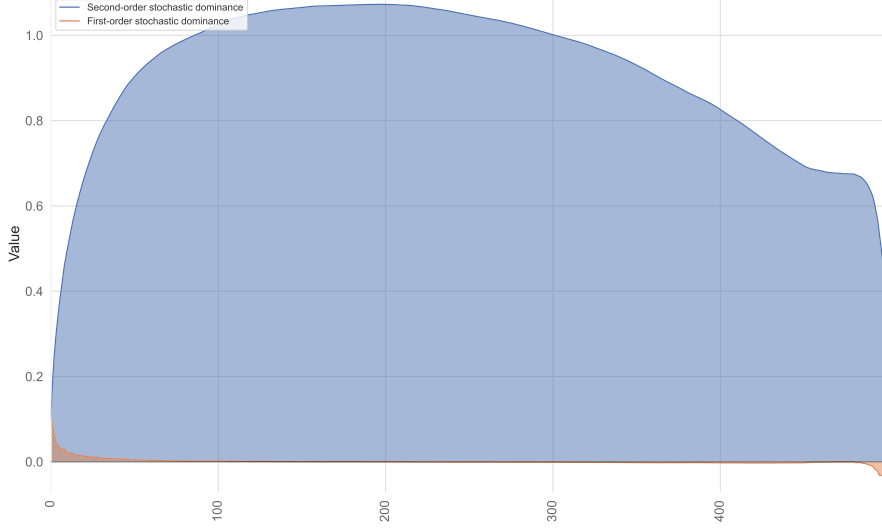


Figure 4: In-sample SD relationships obtained via the unscaled model

In my experience, somewhat informal, the scaled models tend to (most of the time) outperform the unscaled models out-of-sample in terms of excess returns. Unscaled portfolios tend to have smaller drawdowns.

My main idea for future work is to reshape the SSD and FSD relationships between the in-sample portfolios and the benchmark. I would like to try and find portfolios that are more likely to outperform out-of-sample. I thought of three ways of doing this:

- The simplest way would be to play with the scaling factor. The current scaled model has some nice theoretical justification as stated in Fábíán et al. (2011a), but at first I do not have the pretense of connecting any experiment to risk theory. Rather, I would focus entirely on the enhanced indexation aspect when choosing suitable scaling factors. One idea is to give higher importance to the tails. For instance, the current scaling values range from $\frac{1}{S}, \dots, \frac{S}{S}$. Alternatively, we could change it to $\frac{S/2}{S/2}, \frac{S/2-1}{S/2}, \dots, \frac{2}{S/2}, \frac{1}{S/2}, \frac{1}{S/2}, \frac{2}{S/2}, \dots, \frac{S/2-1}{S/2}, \frac{S/2}{S/2}$. With these values, the middle of the distribution would get the lowest emphasis. Not sure if this would help, but I would expect somewhat U-shaped domination charts, with higher tails. There might be other smart ways to define the scaling factor.
- The SD relation compares the sorted returns. When we sort the returns of both the portfolio and the benchmark independently, they lose their correlation: the worst portfolio return is not necessarily at the same period as the worst benchmark return, etc. I am not sure if this is necessarily an issue we should worry about, but an approach to partially mitigating it is to split the in-sample period into K different periods, and find a portfolio that maximises the minimum value of all K SSD relationships.

This variation of the problem can be easily modelled, with a minor adaptation to the separation algorithm. Let the S scenarios be split into sets $S^k, k = \{1, \dots, K\}$. If $S = 200$ and $k = 2$, $S^1 = \{1, \dots, 100\}$ and $S^2 = \{101, \dots, 200\}$ represent a valid reconfiguration. Let r_i^{pk} the i -th return of the portfolio in subset S^k , i.e., if $S^k = \{401, \dots, 500\}$, r_2^{pk} refers to return

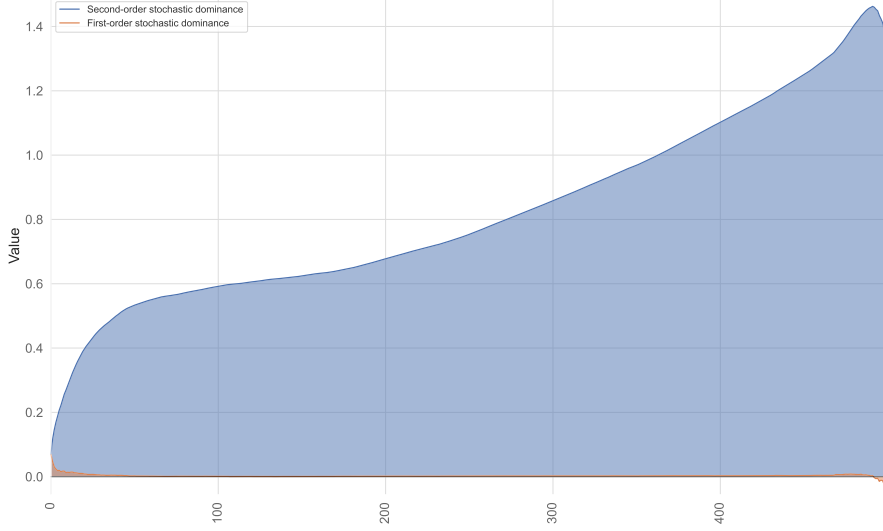


Figure 5: In-sample SD relationships obtained via the scaled model

402. Let r_i^{Ik} be the equivalent definition for the index. In this idea, we want to maximise the minimum of all the following SSD relations at once:

$$\sum_{z=1}^{|S^k|} r_{(z)}^{pk} - r_{(z)}^{Ik} \quad \forall s = \{1, \dots, |S^k|\}, k = \{1, \dots, K\}$$

Let $\hat{\tau}^k = (\hat{\tau}_1^k, \dots, \hat{\tau}_{|S^k|}^k) = (\text{Tail}_{\frac{1}{|S^k|}} R^{Ik}, \dots, \text{Tail}_{\frac{|S^k|}{|S^k|}} R^{Ik})$, so the ordered cumulative returns of the index in the subset $S^k \subseteq S$. Let $\mathcal{V}_k$ be auxiliary variables representing the minimum of the SSD relation between portfolio and benchmark in the subset S_k . The following formulation models this variation of the SSD problem:

$$\max \quad \mathcal{V} \tag{9}$$

$$\text{subject to} \quad \mathcal{V} \leq \mathcal{V}_k \tag{10}$$

$$\frac{s}{|S^k|} \mathcal{V}_k \leq \frac{1}{|S^k|} \sum_{j \in \mathcal{J}_s} \sum_{i=1}^N r_{ij} w_i - \hat{\tau}_s \quad \forall \mathcal{J}_s \subset \{S_1^k, \dots, S_{|S^k|}^k\}, \tag{11}$$

$$|\mathcal{J}_s| = s, s = \{1, \dots, |S^k|\}$$

$$\sum_{i=1}^N w_i = 1 \tag{12}$$

$$\mathcal{V} \in \mathbb{R} \tag{13}$$

Note that the model above is scaled, but it could also be unscaled. The change to the separation algorithm is simple: just check each $\mathcal{V}_k$ separately for violation, instead of checking $\mathcal{V}$. We may add one violated inequality per subset S_k .

Here the sets could be mixed. For instance, $S^1 = \{1, 3, \dots, 497, 499\}$ and $S^2 = \{2, 4, \dots, 498, 500\}$. A possible use for a mixed approach could be for independent SSD optimisation for each market regime. The hypothesis is that the market is not governed by a single statistical distribution, but rather by multiple distributions that alternate over time: for instance, one that best represents bull markets and another that best represents bear markets. A famous algorithm for regime identification is based on Hidden Markov Chains, which can be configured to split historical data into two sets, one with the days where the best fitted distribution is the one for bear markets and another where the best fitted distribution is the one for bull markets. We then could try to optimise two independent SSD relations, one for bear markets and another for bull markets.

Note that if we have 500 sets, each set containing a single return, then we have a model that finds the minimum value in the FSD relation.

- The third approach is the most challenging. I would like to try and find a mixed formulation between SSD and FSD, which would be a generalisation of the unscaled/scaled SSD formulation shown earlier. The idea is to break down the SSD relationship into two or more “dependent” relations (as opposed to independent, as before). For instance, assume we have 500 **ordered** scenarios (not chronological). We split them into k ordered sets S^k . For example, if $k = 2$ we could have $S^1 = \{(1), \dots, (400)\}$ and $S^2 = \{(401), \dots, (500)\}$. Alternatively, the sets could be intermingled, such as $S^1 = \{(1), (3), \dots, (497), (499)\}$ and $S^2 = \{(2), (4), \dots, (498), (500)\}$. We then try to find the portfolio that maximises the minimum of the following SSD relations:

$$\sum_{z=S_1^k}^{S_s^k} r_{(z)}^p - r_{(z)}^I \quad \forall s = \{1, \dots, |S^k|, k = \{1, \dots, K\}$$

To illustrate, consider the mixed example above ($S^1 = \{(1), (3), \dots, (497), (499)\}$ and $S^2 = \{(2), (4), \dots, (498), (500)\}$). The relation for S^1 compares:

$$\begin{aligned} & r_{(1)}^p - r_{(1)}^I \\ & (r_{(1)}^p + r_{(3)}^p) - (r_{(1)}^I + r_{(3)}^I) \\ & (r_{(1)}^p + r_{(3)}^p + r_{(5)}^p) - (r_{(1)}^I + r_{(3)}^I + r_{(5)}^I) \\ & \vdots \end{aligned}$$

Likewise, the relation for S^2 compares:

$$\begin{aligned} & r_{(2)}^p - r_{(2)}^I \\ & (r_{(2)}^p + r_{(4)}^p) - (r_{(2)}^I + r_{(4)}^I) \\ & (r_{(2)}^p + r_{(4)}^p + r_{(6)}^p) - (r_{(2)}^I + r_{(4)}^I + r_{(6)}^I) \\ & \vdots \end{aligned}$$

When I mentioned “dependent” above, I meant that it is still a single SD relation, but stronger than SSD, in the sense that a positive minimum in an hypothetical formulation of the problem above would also ensure SSD dominance.

I thought of trying to formulate this problem as a linear or (probably) integer programming problem. The formulation should be flexible enough to work with any combination of sets. For instance, if we have 500 sets, with $S_1 = \{1\}, \dots, S_{500} = \{500\}$, we would have a model that optimises the problem with relation to FSD. If we have a single set with all scenarios, we have the original formulation. I envision that the scaling factor would also be applicable to this generalised hypothetical formulation.

Part of the research focus would be on finding sets that yield desirable portfolios. For instance, multiple small sets close to the tails, and a larger set in the middle of the distribution. While this all makes sense to me, I am of course open to criticism and alternative thoughts. I have not attempted to model this problem yet.

Other alternative line of research is to study the features of in-sample portfolios and propose exogenous constraints such as to drive the in-sample portfolios to “look” more desirable. For that, I am currently developing a few more in-sample and out-of-sample charts for easier visualisation and understanding of the selected portfolios. Those include CDFs, PDF and cumulative in-sample returns. The latter could help us visualise hypothetical drawdown. From these charts, we might get insights on perceived weaknesses, and then proceed to tackle these with the addition of constraints (for instance, for limiting drawdown).

Something I have not mentioned to you yet is that I have developed quite a framework in Python for testing research ideas in a financial backtesting environment. At some point I can present/describe it to you as I believe it could be extremely helpful to implement and test research hypotheses. Everything I am doing, I am doing as part of this framework.

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